

# Consumer Behavior Decision Intelligence Lab

A Multi-Dimensional Transactional Behavioral Modeling & Decision-Support System

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## 1. Executive Summary

Enterprise transactional ledgers record granular operational sales entries—invoice numbers, product codes, timestamps, units, and price points—but conceal underlying account purchasing behavior. In raw sales files, recurring wholesale commercial accounts appear identical in schema to one-off retail consumers. Standard aggregate reporting answers what products moved, but fails to reveal account-level concentration, order cadence decay, basket drift, or churn risk. The **Consumer Behavior Decision Intelligence Lab (CBDIL)** establishes an end-to-end analytical architecture that bridges raw transaction logs and commercial decision-support. Built on two years of longitudinal transaction records from the official UCI Online Retail II repository (1,067,371 rows), the system implements four-tier event precedence classification, multi-schema PostgreSQL modeling, five-dimensional customer feature engineering, silhouette-optimized behavioral clustering, longitudinal lifecycle state machines across 98,557 customer-months, and six deterministic decision signals. All analytical conclusions are auditable and supported by 229 automated verification tests.

| Metric Dimension             | Quantified Value     | Empirical Context / Significance                                     |
|------------------------------|----------------------|----------------------------------------------------------------------|
| Raw Source Transactions      | 1,067,371 rows       | Two continuous longitudinal years (Dec 2009 – Dec 2011)              |
| Cleaned Staging Records      | 1,033,034 rows       | Post composite-key deduplication (removed 34,337 duplicate lines)    |
| Validated Purchase Lines     | 779,423 lines        | Isolated from 19,494 formal cancellations and 3,462 reversals        |
| Unique Customer Accounts     | 5,878 accounts       | Distinct identified commercial and retail buyer entities             |
| Eligible Behavioral Cohort   | 4,023 accounts       | Sufficient history filter: $\geq 2$ orders and $\geq 90$ days tenure |
| Cumulative Valid Spend       | £17,374,252.42       | Gross purchasing value across 36,969 validated order baskets         |
| Value Concentration          | Top 5% = 53.8% spend | Pareto distribution: top 20% generate 83.1% of eligible volume       |
| Selected Behavioral Segments | K = 3 Segments       | Silhouette score = 0.2749, satisfying min 5% cluster share rule      |
| High-Value Softening Risk    | 261 accounts         | Top spend quintile with $\geq 25\%$ drop in recent 90-day spend      |

## 2. New Contributions and Improvements

CBDIL departs substantially from generic customer segmentation tutorials and standard sales dashboards. The project introduces specific methodological, behavioral, and architectural enhancements:

| Capability / Dimension    | Standard Baseline Approach                                         | CBDIL Analytical Contribution & Systematic Improvement                                                                                                                 |
|---------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event Classification      | Cancellations and returns are deleted blindly or mixed into sales. | Enforces explicit 4-tier precedence (CANCELLATION > REVERSAL > VALID_PURCHASE > INVALID), capturing negative transactions without deflating historical order velocity. |
| Behavioral Dimensions     | Compresses customer behavior into a single composite RFM score.    | Constructs a 5-dimensional behavioral representation: Value, Activity, Assortment Breadth, Interval Stability, and 90-day Momentum.                                    |
| Customer Eligibility      | Single-order or brand-new accounts deleted to fit models.          | Flags accounts with insufficient history (31.5% of population) via metadata flags, maintaining population auditability while isolating unsupervised clustering.        |
| Model Selection Rule      | Cluster count selected subjectively by visual inspection.          | Automated selection rule: smallest K whose silhouette score is $\geq 90\%$ of peak, subject to a minimum 5% cluster share constraint.                                  |
| Longitudinal Dynamics     | Static single-point-in-time lifetime snapshots.                    | Constructs 98,557 monthly customer snapshots across 25 months with a 5-state lifecycle engine (Reactivated > Emerging > Dormant > Softening > Engaged).                |
| Actionable Decision Layer | Descriptive summaries without operational next steps.              | Evaluates 6 deterministic decision signals linked to testing playbooks with 20% holdout control designs and guardrail metrics.                                         |
| Governance & Audit        | Validation performed ad-hoc or omitted entirely.                   | Continuous audit layer tracking 6 systematic data quality controls directly inside the reporting layer with PASS/WARNING status.                                       |

## 3. Business Problem & Analytical Workflow

Commercial distributors face significant operational vulnerability when revenue is heavily concentrated among wholesale clients. Because wholesale ordering cycles are naturally lumpy, relying on historical spend totals masks whether a key account is thriving, stagnating, or actively defecting to a competitor. CBDIL establishes a structured analytical progression:

**Raw Transactions** (1.06M rows) → **Data Quality & Deduplication** (Class 1-4) → **Relational Dimensional Modeling** (Fact Order & Fact Transaction) → **5-Dimensional Feature Engineering** (Value, Activity, Breadth, Stability, Momentum) → **Behavioral Segmentation** (K=3 Cluster Profiles) → **Lifecycle State Machines** (98.5k snapshots) → **Decision Signals** (261 softening accounts flagged) → **Interactive Workbenches** (Streamlit & Power BI).

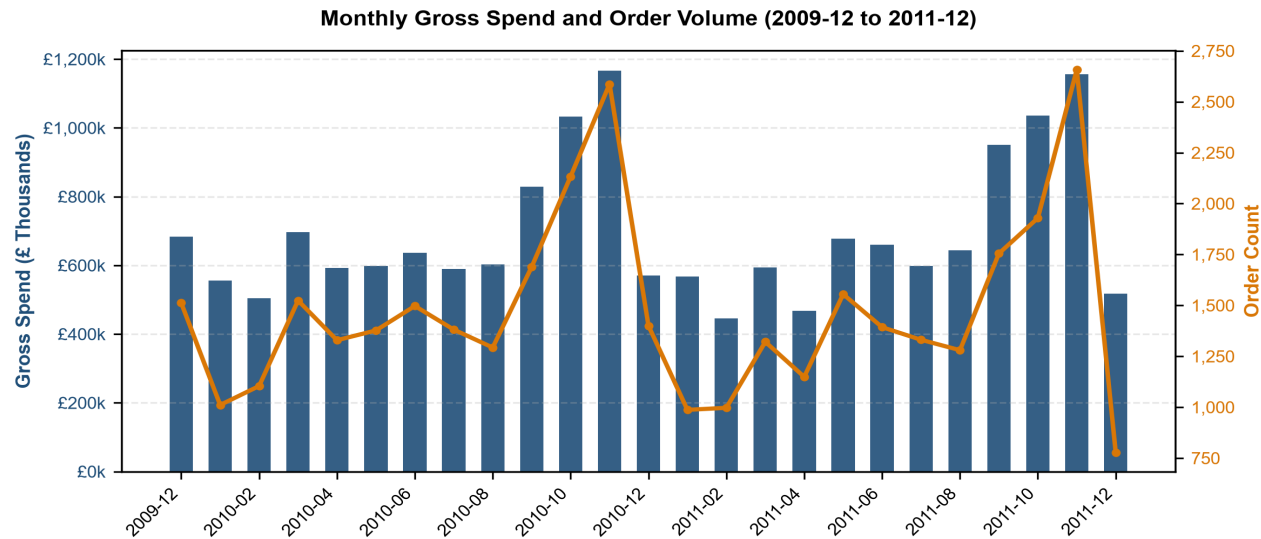
#### 4. Data Quality, Event Classification & Governance

Profiling identified significant data quality anomalies that would severely distort naive calculations. The cleaning pipeline resolves these through deterministic, auditable transformations without modifying raw inputs:

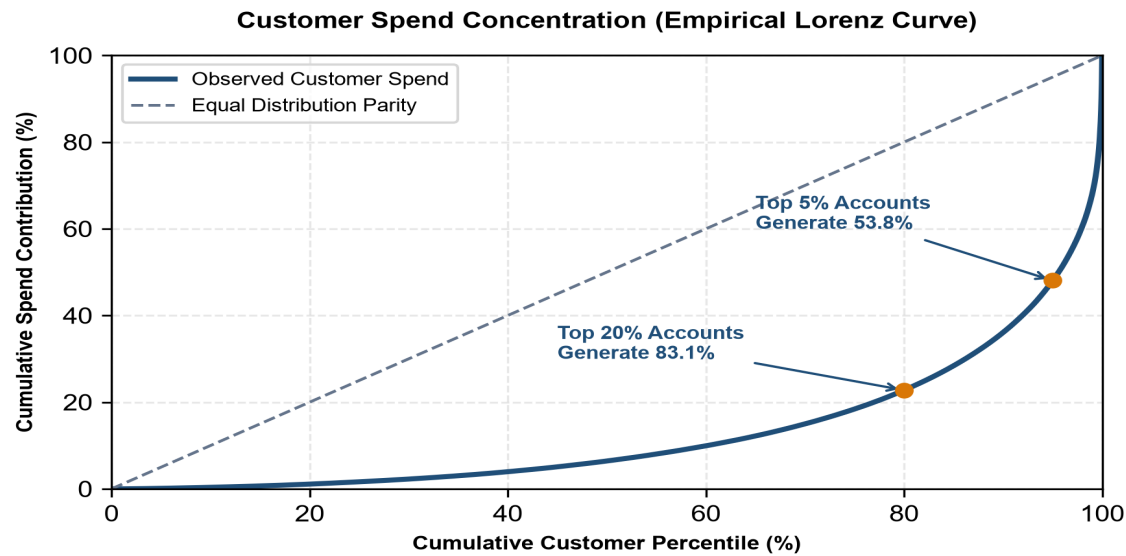
| Data Quality Issue         | Affected Rows | Share  | Analytical Risk                         | Implemented Pipeline Treatment                                                                       |
|----------------------------|---------------|--------|-----------------------------------------|------------------------------------------------------------------------------------------------------|
| Unattributed Rows          | 243,007       | 22.77% | Cannot link to customer lifecycle       | Assigned Class 4 (INVALID_OR_UNUSABLE); excluded from customer features but audited in gross sales.  |
| Duplicate Lines            | 34,337        | 3.22%  | Overcounts item quantities and spend    | Deduplicated via composite hash on invoice, SKU, customer, date, quantity, and unit price.           |
| Order Cancellations        | 19,494        | 1.83%  | Distorts gross purchase order frequency | Classified as Class 1 (CANCELLATION) via 'C' prefix; isolated to compute account-level return rates. |
| Non-Cancellation Negatives | 3,462         | 0.32%  | Distorts sales value with damaged goods | Classified as Class 2 (REVERSAL_OR_RETURN); separated from gross purchase totals.                    |
| Zero / Negative Prices     | 6,207         | 0.58%  | Samples & bad-debt accounting entries   | Classified as Class 4; filtered prior to commercial value modeling.                                  |

#### 5. Macro Purchasing Dynamics & Concentration Analysis

Analysis of gross spend and order volume across the 24-month observation window reveals strong commercial seasonality and extreme revenue concentration:



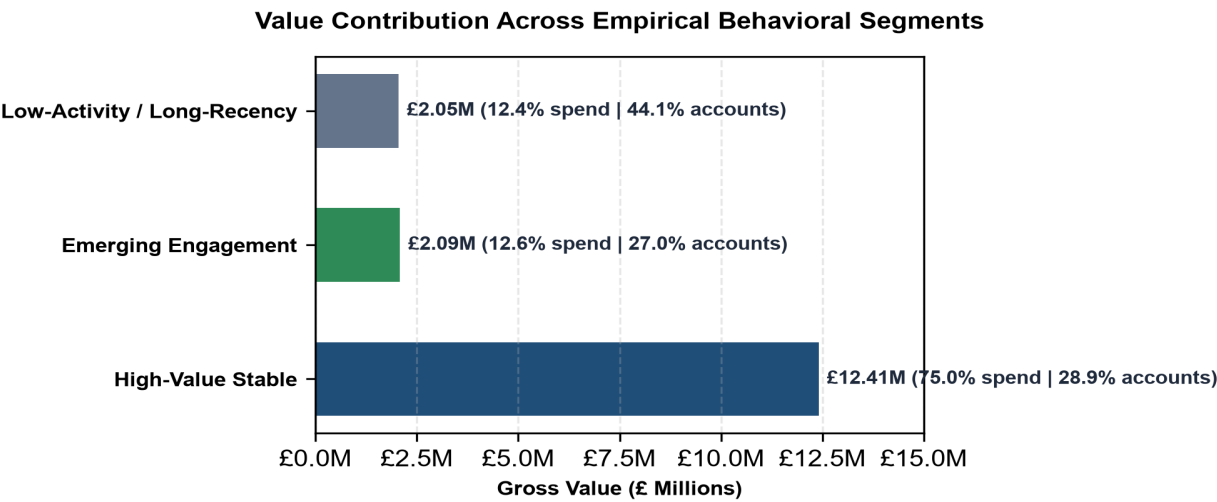
**Figure 1: Monthly Gross Spend and Order Volume Trend (2009-12 to 2011-12).** What it shows: Discrete monthly transaction value (bars, left axis) and total invoice counts (line, right axis). Insight: Strong annual Q4 seasonality peaking in November (£1.46M in 2010 and £1.51M in 2011) reflecting commercial wholesale holiday inventory stocking, followed by an operational contraction in January. Actionable takeaway: Purchasing slowdowns in Q1 represent expected distributor replenishment lulls, whereas decelerations in Q3/Q4 signal abnormal account attrition.



**Figure 2: Customer Spend Concentration (Empirical Lorenz Curve).** What it shows: Cumulative customer population percentile vs cumulative gross spend contribution. Insight: Purchasing revenue conforms to an extreme Pareto distribution: the top 1% of accounts drive 26.4% of total spend, the top 5% drive 53.8%, and the top 20% generate 83.1% (£13.75M of £16.55M eligible spend). Actionable takeaway: Financial performance is heavily exposed to a narrow wholesale core, validating the need for dedicated account capacity protection.

6. Behavioral Segmentation Architecture & Cluster Profiles

K-Means clustering was executed on the 4,023 eligible accounts across standardized log-spend, log-orders, recency, log-SKUs, interpurchase gap, and momentum variables. K=3 was selected deterministically based on silhouette maximization (0.2749) and minimum cluster size representation:



**Figure 3: Value Contribution Across Empirical Behavioral Segments.** What it shows: Gross historical spend (£ Millions) and relative spend share across the three identified segments. Insight: The customer base bifurcates sharply: 28.9% of accounts (Segment 2) capture 75.0% of cumulative revenue (£12.41M), while 44.1% of accounts (Segment 1) account for only 12.4% (£2.05M) despite their numerical dominance.

| Segment Name                | Account Count | Spend Share     | Median Spend | Median Orders | Median Recency | Median SKUs | Strategic Inquiry                                                              |
|-----------------------------|---------------|-----------------|--------------|---------------|----------------|-------------|--------------------------------------------------------------------------------|
| High-Value Stable           | 1,164 (28.9%) | £12.41M (75.0%) | £4,967.17    | 13 orders     | 23 days        | 173 SKUs    | What delivery terms and inventory SLAs are required to protect these accounts? |
| Emerging Engagement         | 1,084 (27.0%) | £2.09M (12.6%)  | £1,259.86    | 4 orders      | 29 days        | 65 SKUs     | Which category affinity bundles can accelerate basket adoption in this cohort? |
| Low-Activity / Long-Recency | 1,775 (44.1%) | £2.05M (12.4%)  | £815.69      | 3 orders      | 266 days       | 40 SKUs     | What proportion represents seasonal holiday buyers vs ceased operations?       |

7. Deterministic Decision Signals & Action Strategies

CBDIL converts behavioral metrics into structured decision triggers by evaluating six deterministic, rule-based signals. Each signal carries an objective threshold, an account population, a severity tier, and attached audit limitations:

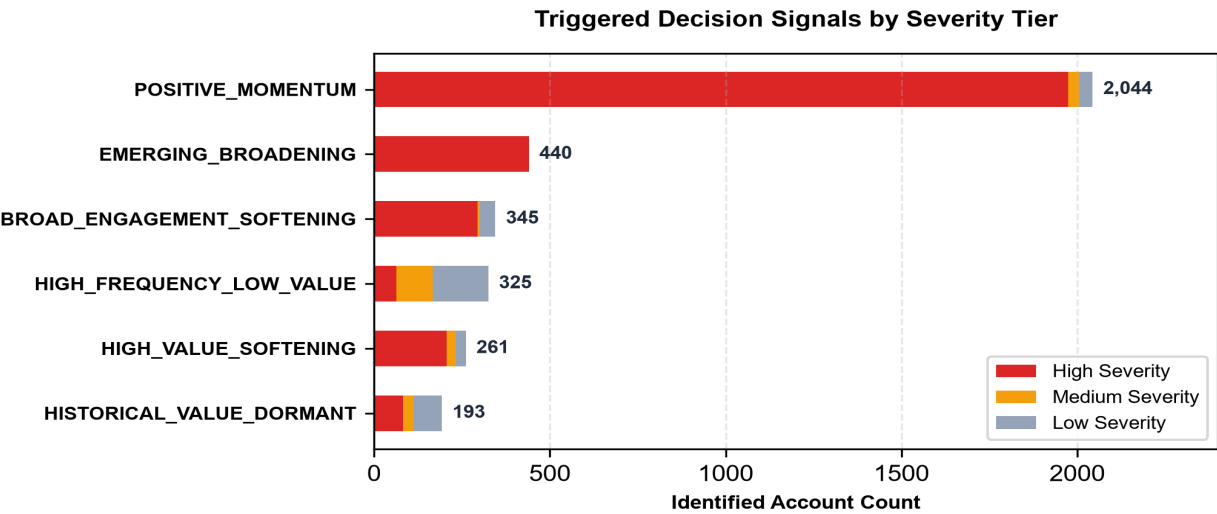


Figure 4: Triggered Decision Signals by Severity Tier. What it shows: Volume of customer accounts meeting deterministic signal trigger rules, broken down by severity tier (High, Medium, Low). Insight: While positive growth is widespread (2,044 accounts exhibiting positive momentum), severe retention risks are concentrated in 261 accounts flagged under HIGH\_VALUE\_SOFTENING, representing £2.56M in cumulative historical spend.

| Signal Identifier          | Exact Trigger Rule                                   | Accounts | Cumulative Spend | Possible Business Action                                                | Evaluation KPI / Guardrail                   |
|----------------------------|------------------------------------------------------|----------|------------------|-------------------------------------------------------------------------|----------------------------------------------|
| HIGH_VALUE_SOFTENING       | Spend >= P80 (£2,910) and Value Momentum <= -25%     | 261      | £2,559,881.18    | Commercial account manager check-in to review pricing and availability. | 90d Spend Recovery % / Return Rate Guardrail |
| HIGH_FREQUENCY_LOW_VALUE   | Orders >= P80 (8 orders) and AOV <= P40 (£233.82)    | 325      | £992,802.24      | Implement minimum order size thresholds or freight incentives.          | Average Order Value (AOV) / Order Frequency  |
| EMERGING_BROADENING        | State = EMERGING and Product Breadth Momentum > 0%   | 440      | £269,788.27      | Present category affinity recommendation bundles during checkout.       | 60d Repeat Conversion / Category Return Rate |
| HISTORICAL_VALUE_DORMANT   | Spend >= P80 (£2,910) and State = DORMANT (>120d)    | 193      | £1,474,863.53    | Evaluate seasonal re-engagement offer with 20% holdout control.         | Reactivation Rate / Discount Margin Erosion  |
| BROAD_ENGAGEMENT_SOFTENING | SKUs >= P70 (85 SKUs) and Frequency Momentum <= -25% | 345      | £2,180,615.61    | Audit catalog stock-outs across core wholesale lines.                   | Frequency Recovery / Order Cancellation Rate |
| POSITIVE_MOMENTUM          | Spend Momentum >= +25% and Freq Momentum >= +25%     | 2,044    | £8,400,789.95    | Guarantee supply-chain capacity and priority inventory allocation.      | OTIF Order Fulfillment / Delivery Delay Rate |

8. Dashboard Architecture & Interactive User Journey

The analytical deliverables are operationalized through two cohesive dashboard interfaces: an interactive Streamlit workbench and a star-schema Power BI semantic reporting model. Both interfaces follow an identical six-page navigation logic:

| Dashboard Page   | Primary User Question Answered                                             | Key Visuals & Interactive Elements                                                  | Decision Supported                                                      |
|------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Page 1: Overview | What is the overall scale and spend concentration of the customer base?    | KPI cards, Monthly spend/order trend, Empirical Lorenz curve, Lifecycle state bars. | Macro business health assessment and key-account volume monitoring.     |
| Page 2: Segments | How do customers separate behaviorally across value, recency, and breadth? | Segment metrics matrix, Recency vs log-spend scatter plot, SKU breadth boxplots.    | Identifying structural differences in wholesale vs retail buyer groups. |

|                            |                                                                                   |                                                                                           |                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Page 3: Deep Dive          | What specific behavioral characteristics define an individual segment?            | Recent vs prior 90-day spend comparison bars, active monthly trend, inquiry cards.        | Formulating tailored merchandising and contract terms for specific segments.          |
| Page 4: Behavior Over Time | Where is customer behavior moving month-over-month across lifecycle states?       | Monthly stacked state area chart, state transition matrix heatmap, cohort retention grid. | Measuring attrition rates from active into softening and dormant states.              |
| Page 5: Decision Signals   | Which specific customer accounts warrant proactive review or capacity protection? | Signal inventory bar chart, severity distribution breakdown, customer drill-down table.   | Isolating high-risk accounts (e.g., 261 softening clients) for targeted outreach.     |
| Page 6: Customer Explorer  | Why is a specific customer classified under a particular segment or signal?       | Customer ID searchable dropdown, diagnostic metrics, classification rationale box.        | Pre-contact audit of individual wholesale client purchasing history and risk factors. |

9. End-to-End Account Walkthrough (Customer #14156)

To verify that the system operates as a connected pipeline rather than isolated scripts, consider the empirical trajectory of Customer #14156:

1. **Raw Ingestion:** Account #14156 places 156 discrete orders across Ireland (EIRE) between Dec 2009 and Nov 2011.

2. **Cleaning & Classification:** Pipeline deduplicates 14 duplicate line entries and classifies 2,610 valid purchase lines and 89 formal cancellation lines.

3. **Feature Calculation:** Total Spend = £313,437.62 (Top 0.1% tier); Orders = 156; Recency = 9 days; Recent 90d Spend = £18,521.29; Prior 90d Spend = £53,838.44; Value Momentum = -65.6% drop.

4. **Behavioral Segmentation:** Clustered into Segment 2: High-Value Stable based on high cumulative spend and catalog reach (450+ unique SKUs).

5. **Lifecycle State Machine:** Classified as SOFTENING as of reference date due to recent spend drop exceeding 25%.

6. **Decision Signal:** Triggers HIGH\_VALUE\_SOFTENING (High Severity) with attached limitation regarding wholesale replenishment timing.

7. **Operational Action:** Linked to Strategy STRAT-01 for proactive commercial outreach, evaluated via a 20% holdout control trial with a 90-day recovery window.

10. System Controls, Limitations & Technical Conclusion

The project incorporates an explicit control layer (‘reporting.data\_control\_summary’) evaluating 6 continuous audit checks. All 229 automated unit and integration tests execute cleanly in under 8 seconds with zero linter errors. However, five methodological boundaries must be respected when interpreting conclusions:

1. **Wholesale Scope:** The merchant represents a commercial giftware distributor; reorder intervals cannot be generalized to consumer banking or SaaS subscriptions.

2. **Observational Bounds:** Marketing interventions are unobserved in the source data; decision signals define hypotheses for testing rather than guaranteed causal uplifts.

3. **Wholesale Purchase Lumpiness:** Inactivity intervals of 60–90 days frequently reflect standard distributor restocking cycles rather than account cancellation.

4. **Unattributed Volume (22.77%):** Guest checkout transactions lacking customer identifiers are isolated from account clustering while preserved in gross financial accounting.

5. **Zero Baseline Discontinuities:** Accounts with zero orders in prior baseline periods are evaluated via deterministic default momentum rates rather than artificial interpolation.

**Technical Conclusion:** The Consumer Behavior Decision Intelligence Lab establishes an auditable, reproducible, and statistically disciplined customer intelligence framework. By transforming raw transactional data into multi-dimensional features, behavioral segments, lifecycle snapshots, and deterministic decision signals, the system equips commercial and analytics teams with transparent tools to monitor risk, protect high-value accounts, and test targeted interventions.